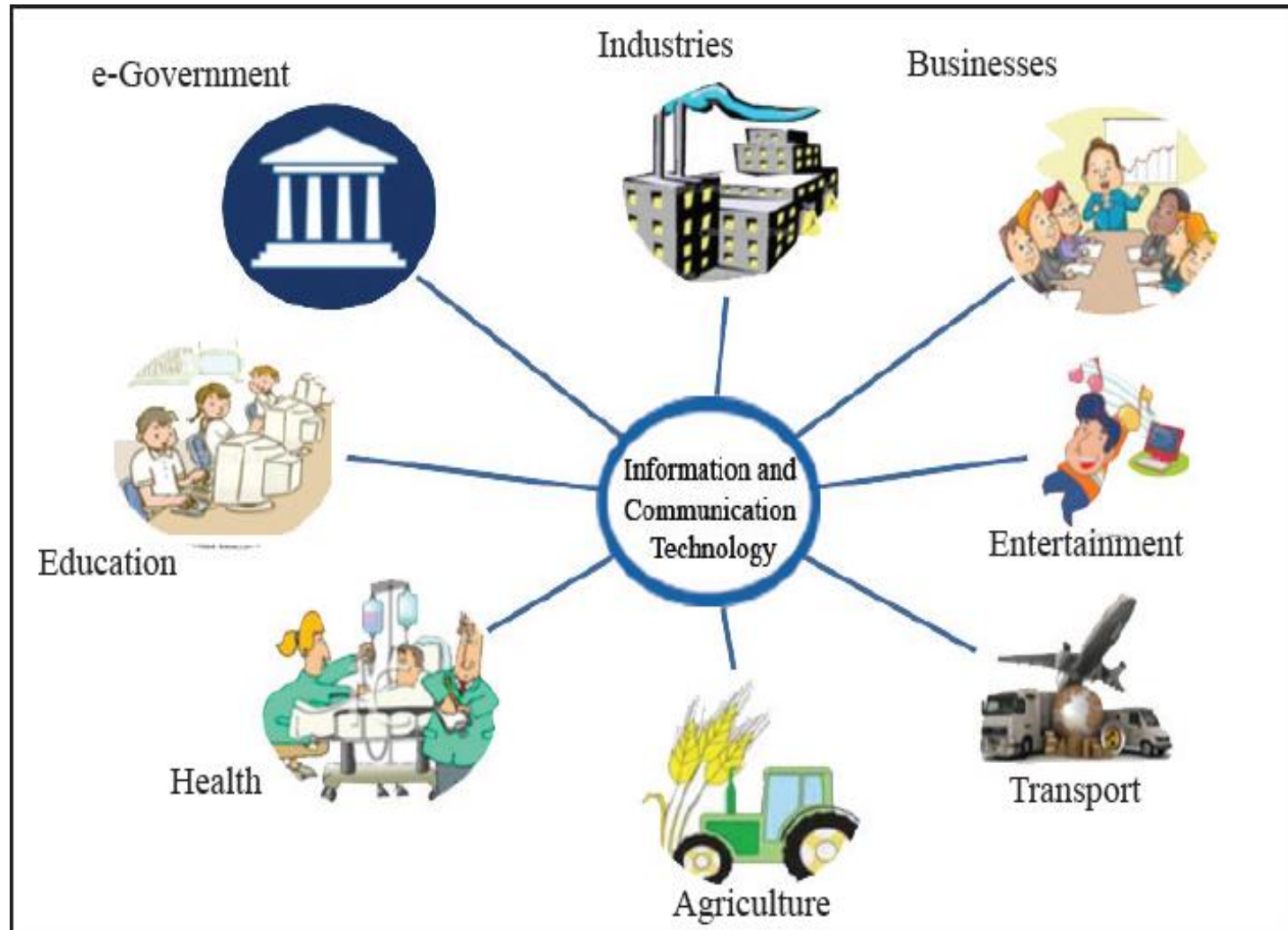


Introduction to ICT & Computer Hardware

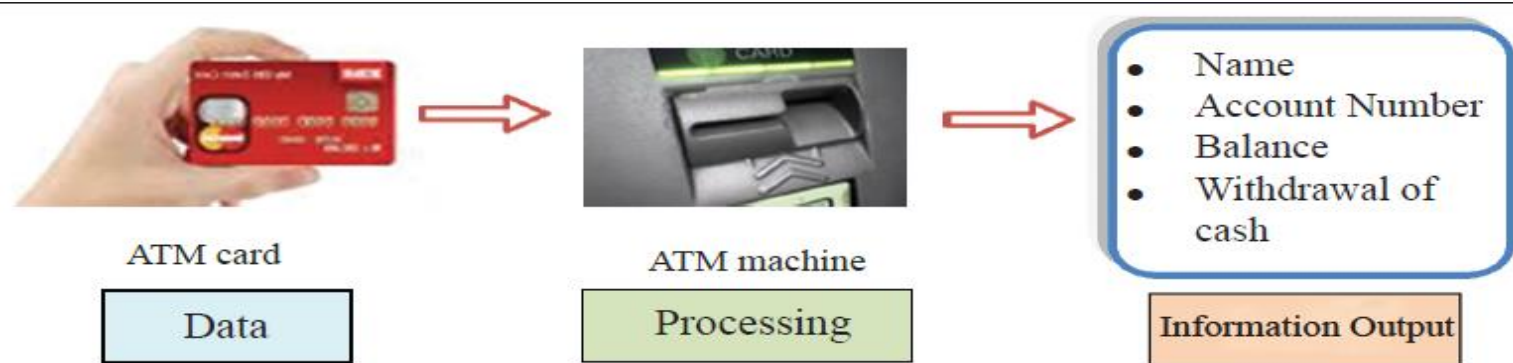


Applications of Information and Communication Technology



Data and Information

- The numbers, words, images and symbols which do not bear a meaning, when standing alone are called data. While we can arrive at meaningful information by arranging and processing data, we can use them to make decisions also.
- Submitting data for processing is called “Input” and the result we get after processing is called “Output.” We can call the collection of all these components above a “System.” Hence the purpose of a system is to receive data, process and store them and provide the results when required.



What is a computer?

- A computer can be described as an electronic device, which accepts or collects data, processes them according to the given instructions and produces the desired output.
- The processing of input into output is directed by the software, but performed by the hardware.



Classification of computers

According to size

Super Computers

Mainframe
Computers

Mini Computers

Personal Computers/ Micro Computers

According to technology

Analog Computers

Digital Computers

Hybrid Computers

Classification according to size

Super Computers



- These computers have higher computing power. They are large in size, expensive and rare. These are used for scientific and engineering functions and to solve complex mathematical problems.
- Example: space projects

Mainframe Computers



- These have relatively less computing power and less expensive than the super computers. In mainframe computers, a number of users get connected to one computer using a number of terminals sharing, saving and retrieving data. These are used in large scale businesses and in e-business.

Mini Computers

- These are slower, less powerful and less expensive than Mainframe computers and are also called common purpose computers. These are easy to use.
- A few users get connected to the central computer here through a few terminals to share data and information. Some banking systems use mini computers.

Personal Computers/ Micro Computers

- This is a small computer meant for personal use. This consists of a small memory capacity, is smaller in size, is less speedy and is less expensive. Further, this consumes little electricity.
- Laptop computers and personal computers are some examples for this category.

Classification according to Technology



Analog Computers



Digital Computers



Hybrid Computers

Analog Computers

- **Analog computers** process **analog** (continuously varying) data.
- **Analog computers** operate on mathematical variables in the form of physical quantities that are continuously varying.
- For **example** temperature, pressure, voltages, ecg machine etc.
- A thermometer is a **simple analog computer**

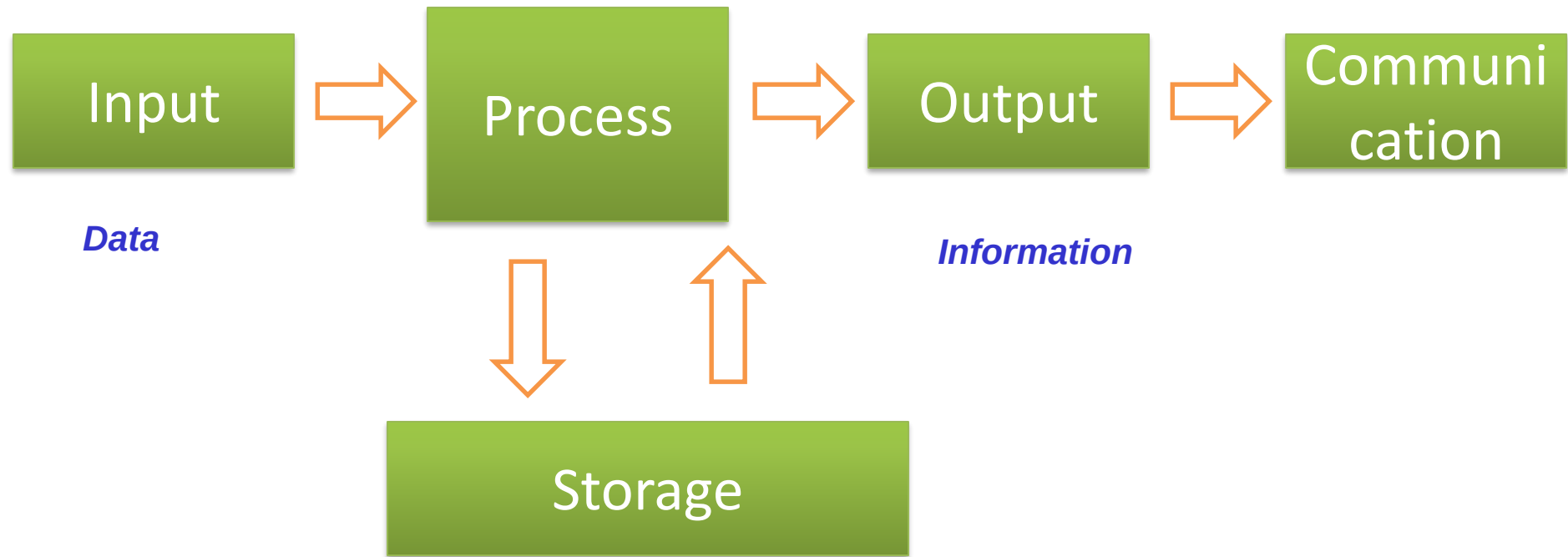
Digital Computers

- A ***computer*** that accepts and processes data that has been converted into binary numbers.
- Example: laptops, computers

Hybrid Computers

- A hybrid computer is a type of computer that offers the functionalities of both a digital and an analog computer.
- It is designed to include a working analog unit that is powerful for calculations, yet has a readily available digital memory.

Computer Concept



Computer Concept

Input

- A computer accept data that is provided by means of an input device.

Process

- A computer performs operations on the data to transform it in some way.

Output

- A computer produces output on a device, that shows the results of processing.

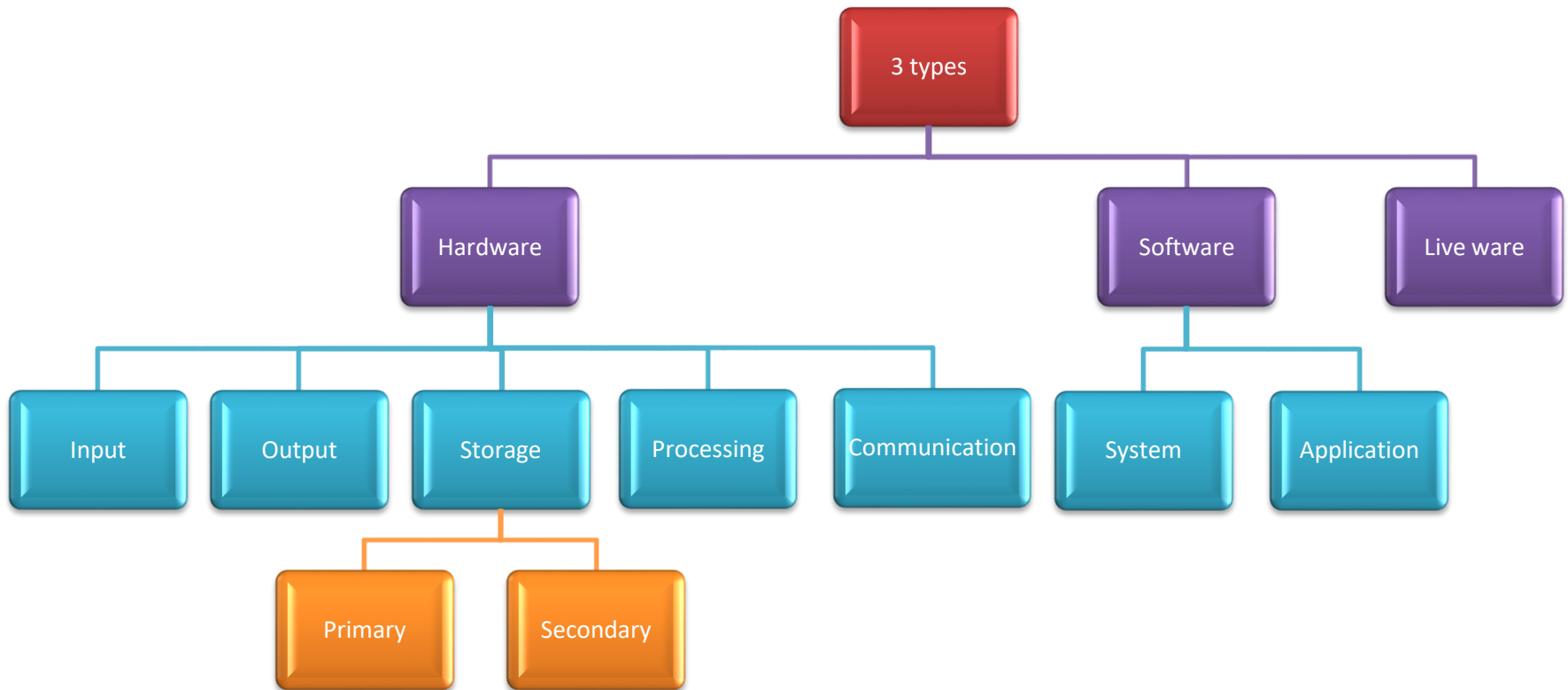
Communication

- Share the output with others

Storage

- A computer stores the results of processing operations for future use.

Components of a Computer



What are Hardware and software?

Hardware

- **Electronic and mechanical equipment**

Software

- **Instructions that tell hardware how to perform tasks**

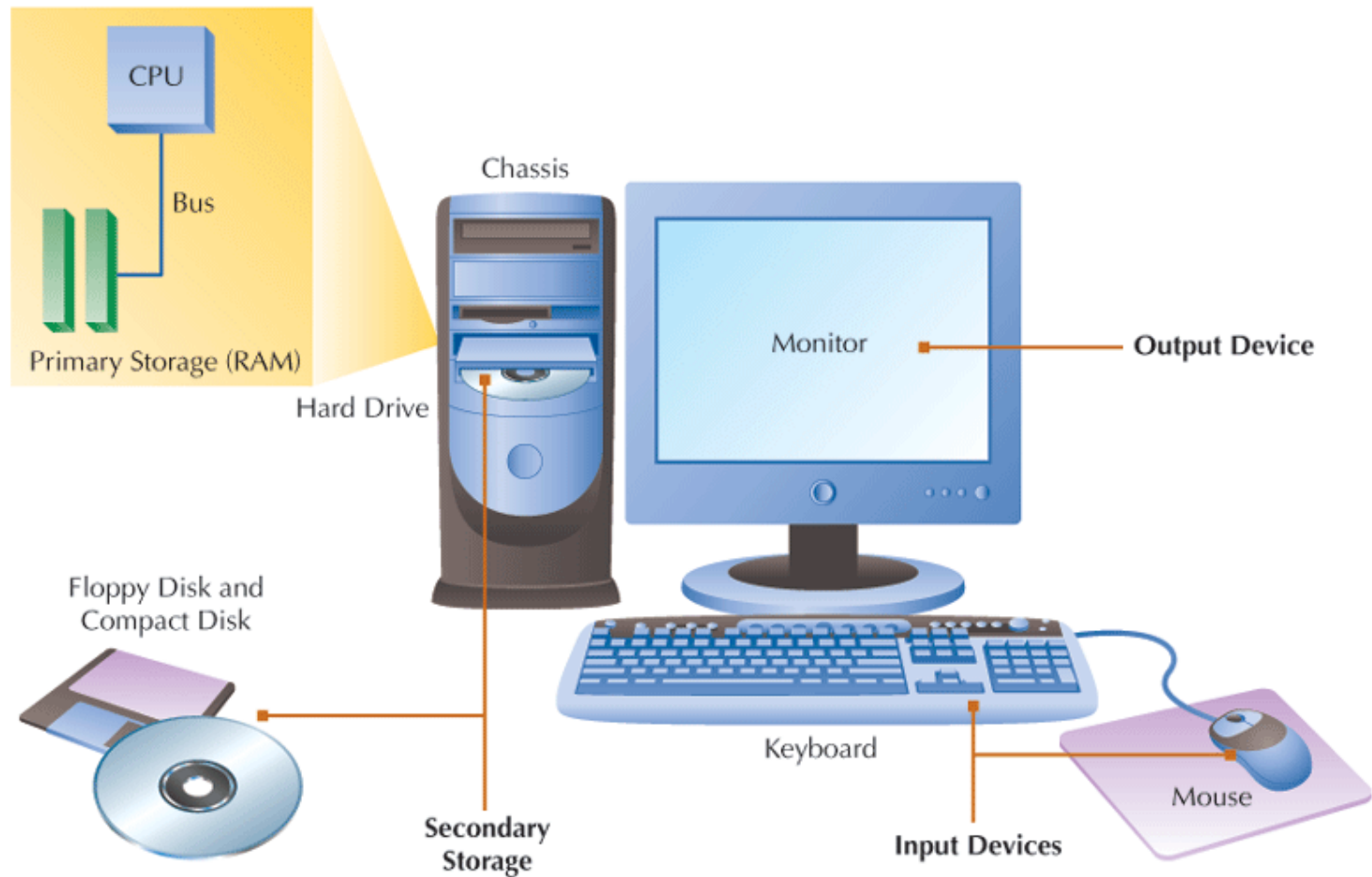
Live ware

- **Live ware is the user who use the Computer**

Computer Hardware

Core Components of a Computer

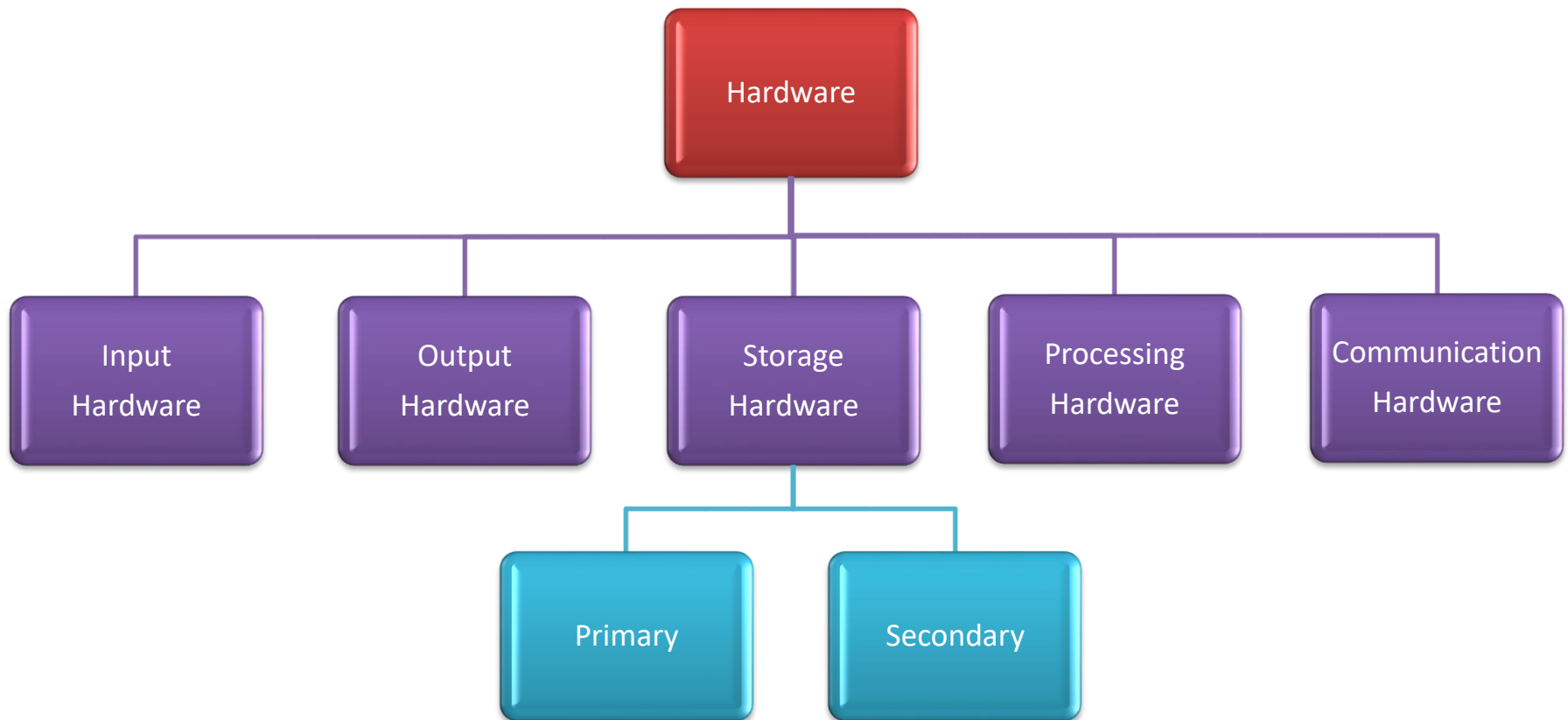
Figure 2.1 Core Components of the Computer



Common Computer Hardware Components?



Components of a Computer



Input Devices



What is an Output Device?

- Hardware that conveys information to a user



Communications Device

- Establishes a connection between two computers using cable, telephone lines, and satellites

Eg: Modem, Switch, Router



Modem

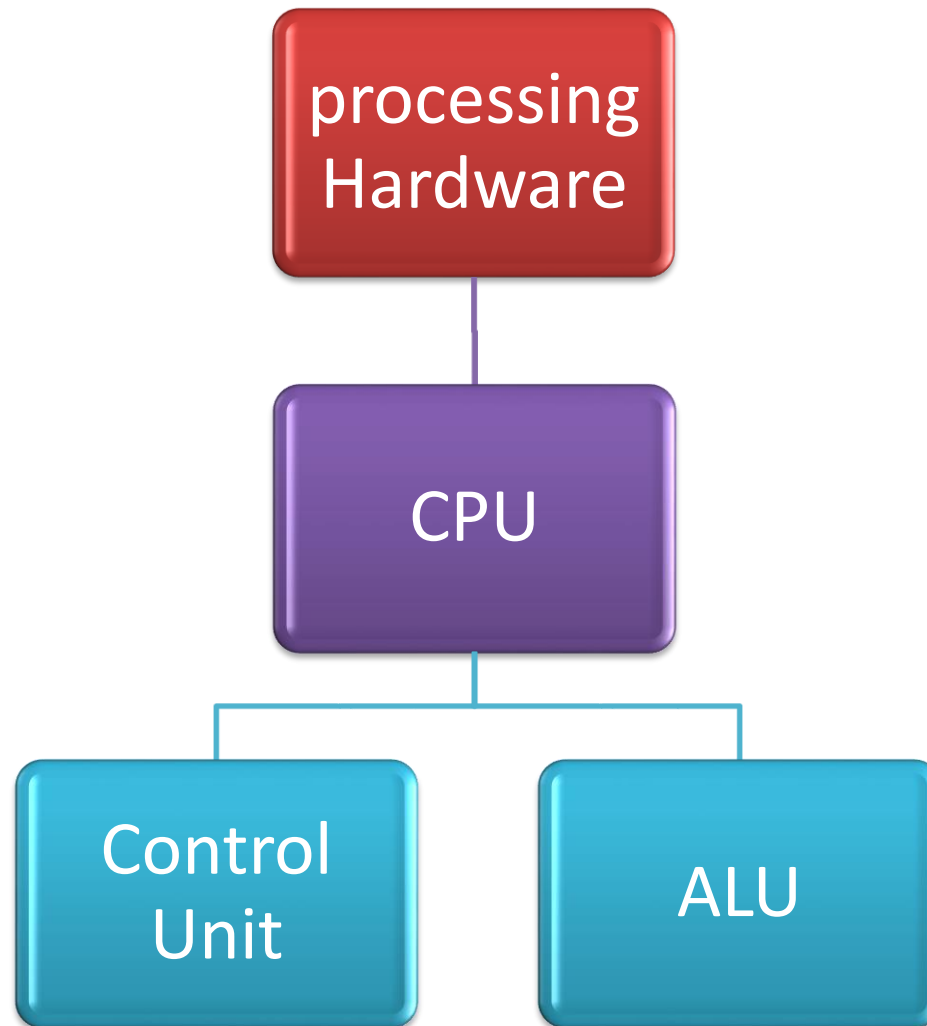


Switch

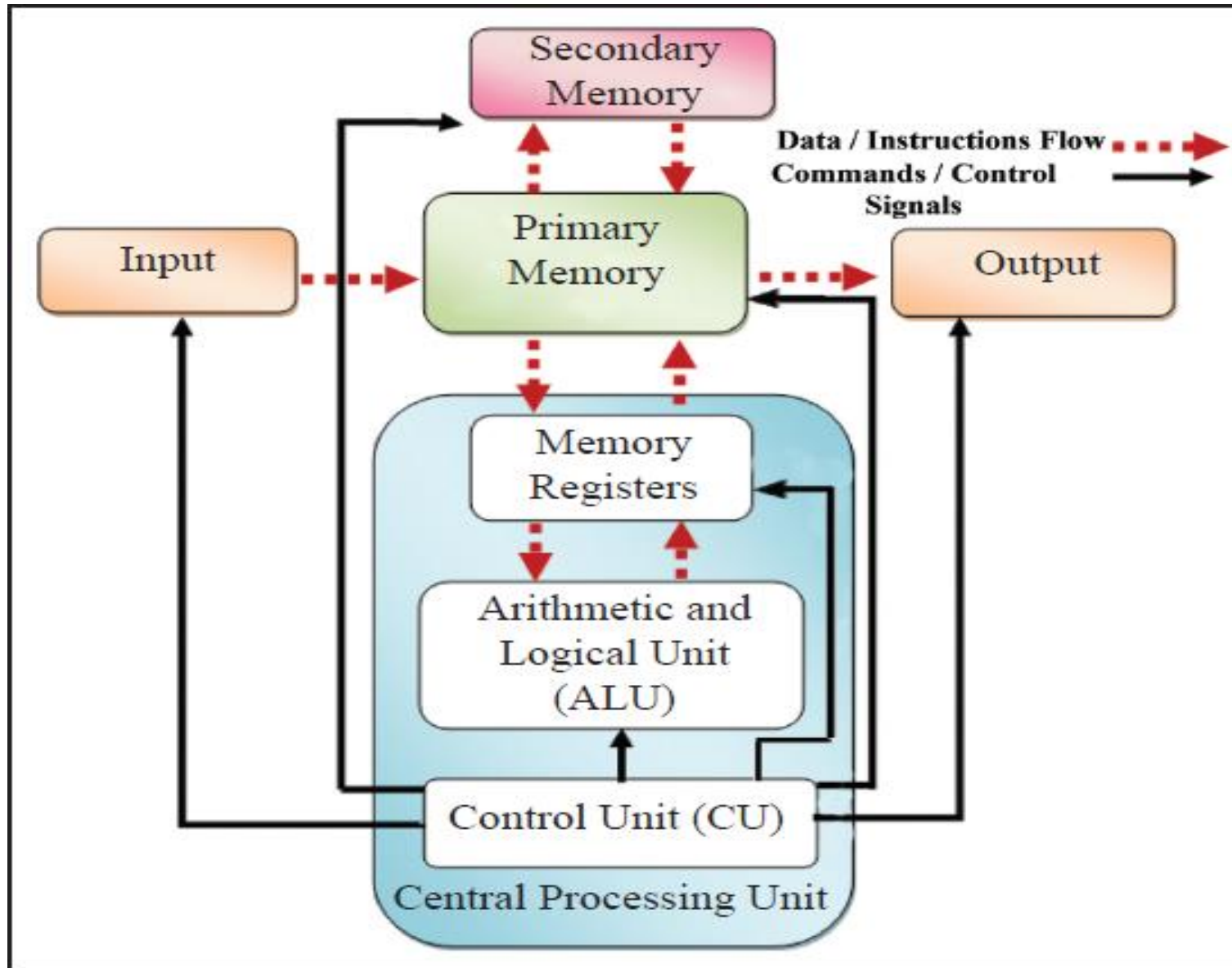


Router

Processing Hardware



Function of a Computer System



CPU (Central Processing Unit)

- CPU is the brain of the computer.
- Also called a processor
- Carries out instructions that tell computer what to do
- Has an internal clock, to synchronize all operations in the machine cycle. The speed is expressed in MHz.
generally the faster the clock speed the faster the computer can process information.
- As one computer generation has involved to the next
 - Size of the CPU has become smaller & smaller
 - Speed & capacity have increased tremendously.

Control Unit (CU)

- Is responsible for directing & coordinating most of the computer system activities.
- It directs flow of data around the CPU and around the computer.
- The CU contains the CPU's instructions for carrying out commands. The instruction set, which is built into the circuitry of the control unit, is a list of all the operations that the CPU can perform.

Arithmetic & Logic unit(ALU)

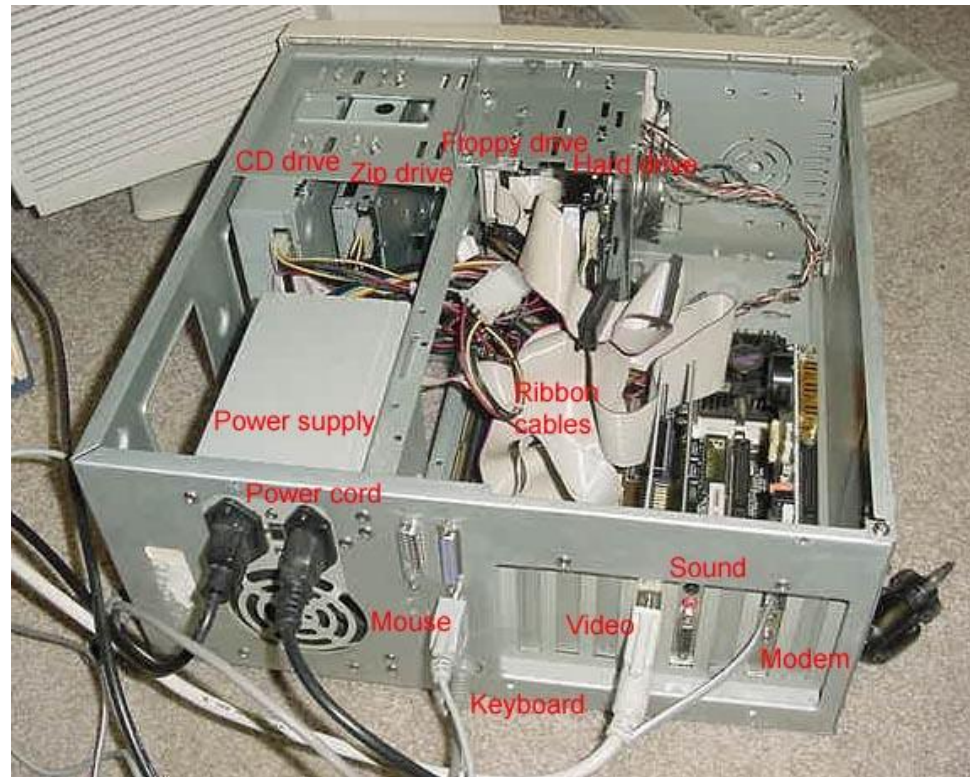
- The ALU performs all the arithmetic & logical (comparison) functions,
 - Adding
 - Subtracting
 - Dividing
 - Multiplying

Memory Registers

- Positioned in the Central Processing Unit.
- Temporarily stores data needed for the function of ALU.

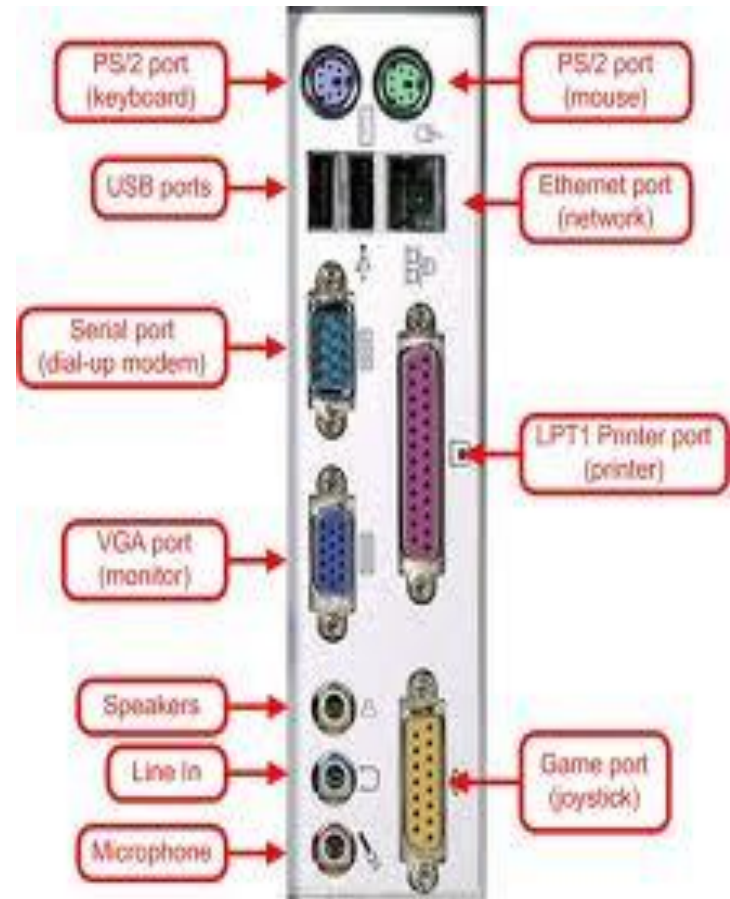
System Unit & *chassis*

- Drives are housed in drive bays which are accessed at the front of the case.
- Internal drives, such as the hard disk drive, are installed in internal bays that are not typically as accessible as the external drives.



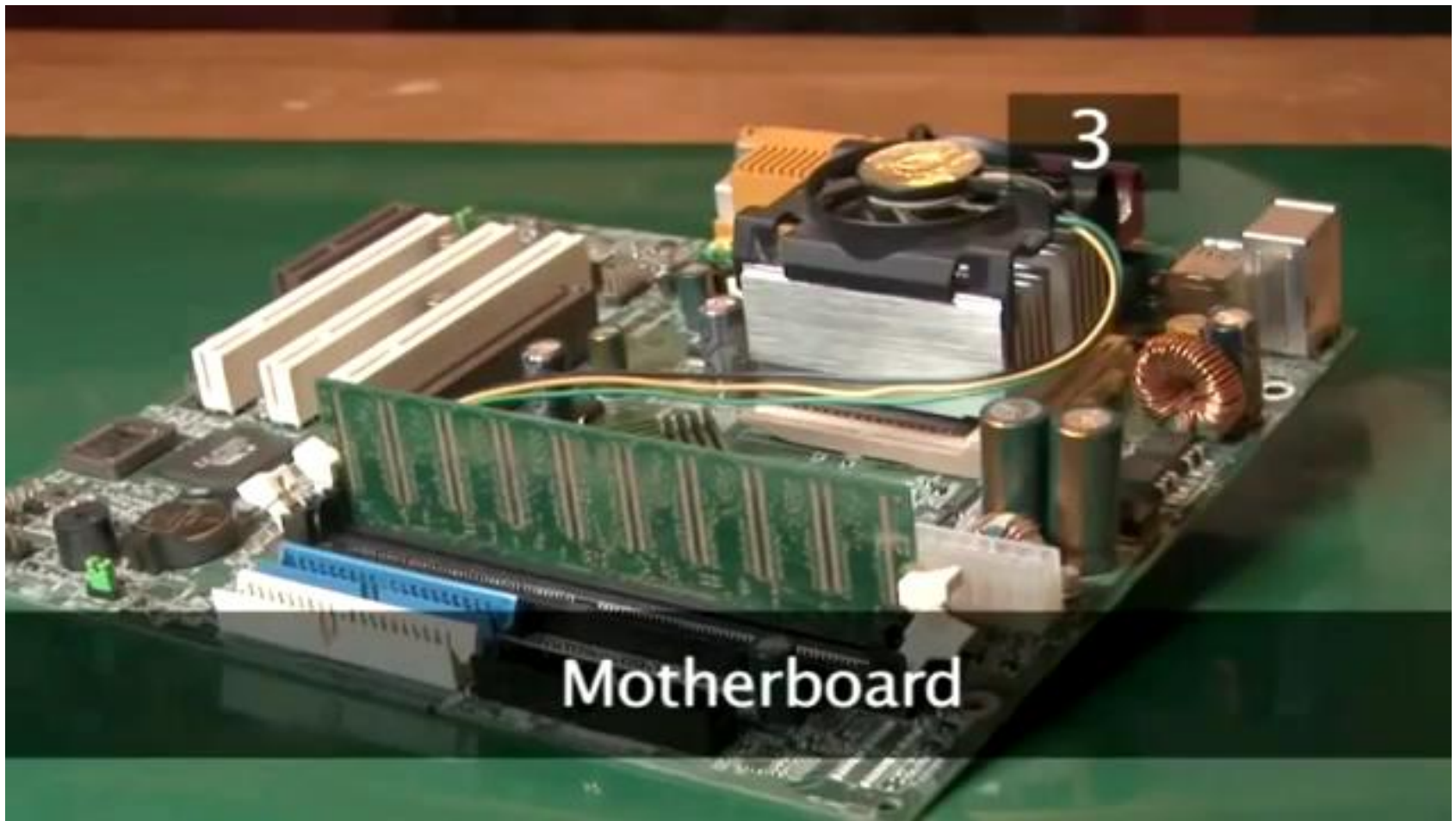
The System Unit

- The System Unit houses the central processing unit, memory modules, expansion slots, and electronic circuitry as well as expansion cards that are all attached to the motherboard; along with disk drives, a fan or fans to keep it cool, and the power supply.
- All other devices (monitor, keyboard, mouse, etc.), are linked either directly or indirectly into the system unit.

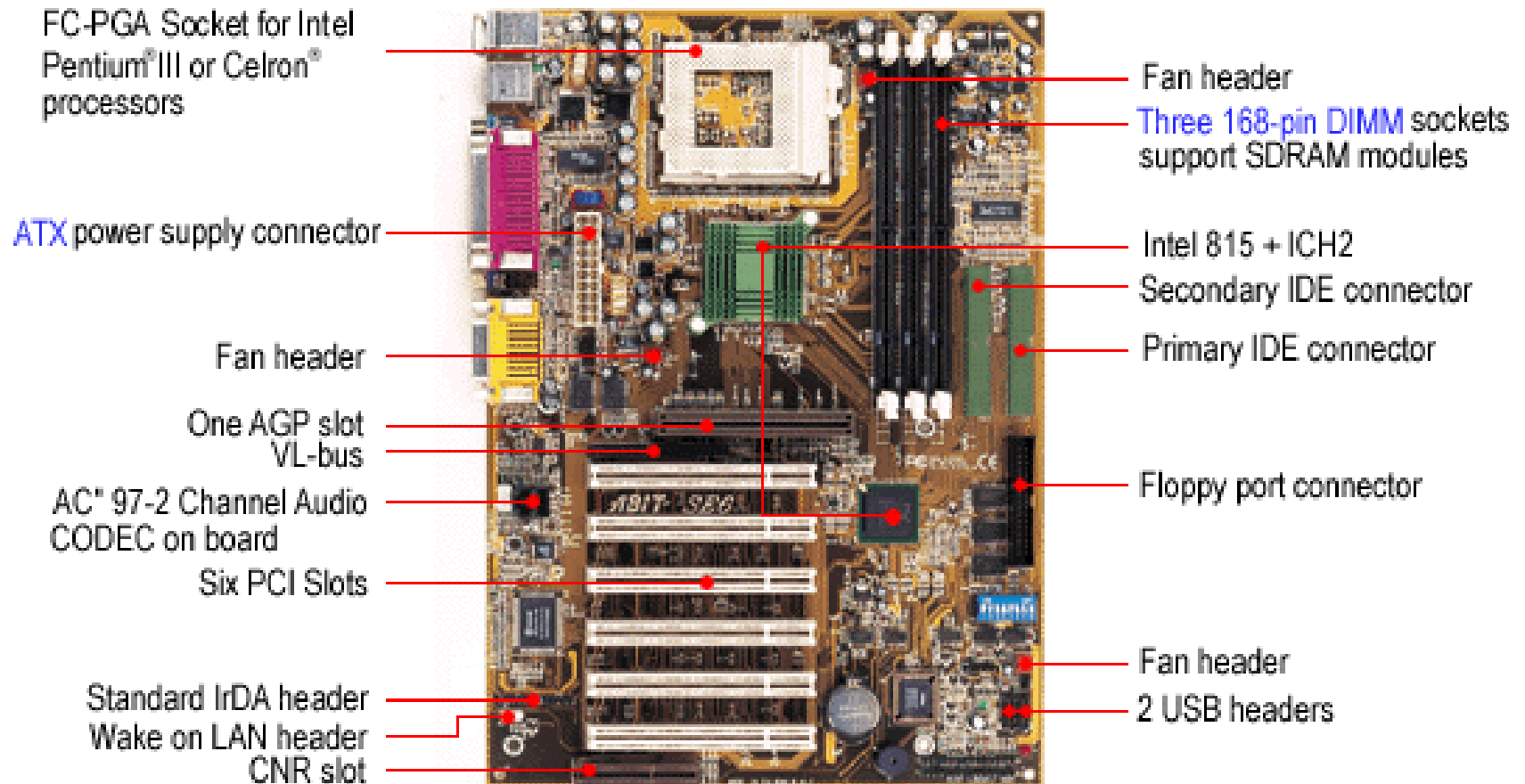


Motherboard

- The mother board is the main circuit board of a microcomputer. It contains:
 - The Central Processing Unit (CPU)
 - The Basic Input/Output System (BIOS)
 - Memory
 - Mass storage interfaces
 - Serial and parallel ports,
 - expansion slots and
 - all the controllers for standard peripheral devices like the keyboard, disk drive and display screen.



Motherboard...



Motherboard...

- The BIOS
 - BIOS stands for *Basic Input/Output System*. It is the lowest-level software in the computer; it acts as an interface between the hardware (especially the chipset and processor) and the operating system.
 - The BIOS provides access to the system hardware and enables the creation of the higher-level operating systems that you use to run your applications.

Motherboard...

- The BIOS...
 - The BIOS is also responsible for allowing you to control your computer's hardware settings, for booting up the machine when you turn on the power or hit the reset button, and various other system functions.

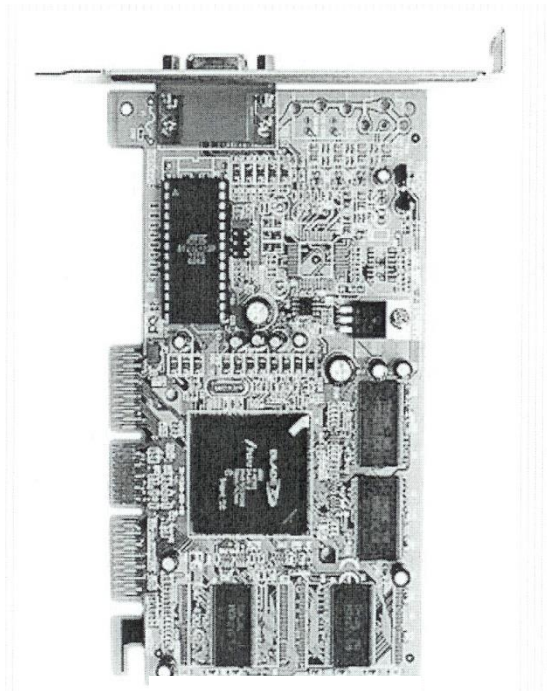
Expansion Slots and Cards

- Expansion slots are sockets to provide direct connections to the common electrical bus, allowing you to insert a circuit board into the motherboard.
- Typical Expansion Cards:
 - Video Cards
 - Sound Cards
 - Modem Cards
 - Network Interface Cards (NIC)
- Laptops and portable computers typically have PC Cards – thin credit-card sized devices used to add memory, disk drives, etc.

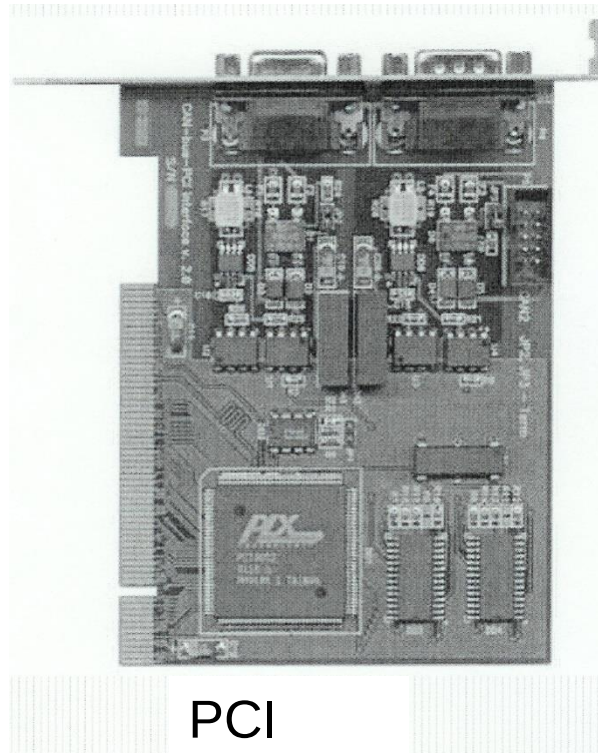
Expansion Slots and Cards...

- For further research about connecting devices to your computer, look up:
 - Peripheral Component Interconnect (PCI)
 - Accelerated Graphics Port (AGP)
 - Universal Serial Bus (USB)
 - Small Computer System Interface (SCSI)
 - Integrated Drive Electronics (IDE)

Expansion Cards



AGP



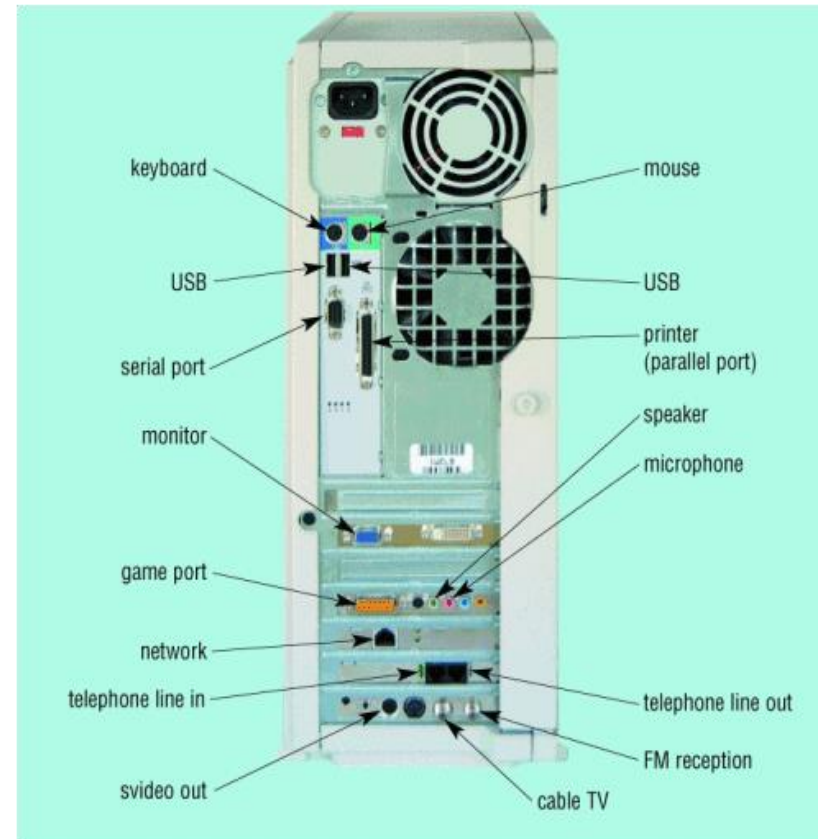
PCI



USB

Ports

- Ports are sockets that allow you to plug in device connectors to access the common electrical bus on the motherboard.
- Ports are usually found on the back of the system unit, but newer styles also have some of them conveniently located on the front.
- Ports allow specific types of connectors (which partly reflects changing technology as well as various kinds of technology).



Serial Port



- A serial port is used to connect external modems or an older computer mouse to the computer.
- There are two serial port versions :
 - 9-pin version , or
 - 25-pin version.
- Serial port is a male port and the typical data transfer rate over a serial port is 115 kilobits per second.

Video Card Port



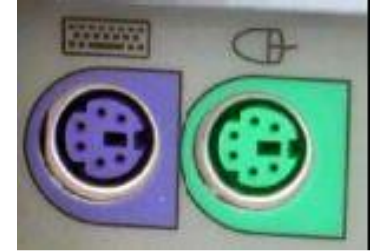
- This connector is used to connect a computer display monitor to a computer's video card.
- The port is a female port and has 15 holes.

USB Port

- A universal serial bus (USB) port is a fairly new port - introduced around 1997.
- This port can be used to connect all kinds of external devices, such as external hard drives, printers and mice, scanners. USB-compliant devices can draw power from a USB port. USB4 connectors were introduced in 2019 and are capable of transferring data at 40 Gbps.



PS/2 Port



- A PS/2 port, also referred to as a mouse port, is used to connect a mouse or keyboard with the computer.
- Most computers come with two PS/2 ports, one for the keyboard and the other for the mouse.

BUS

- In microcomputers the term BUS Refers to the paths between the components of a Computer.
- CPU and other devices connected with buses.
- There are two main buses in a computer
 - Data bus
 - Address bus

Data BUS

- Is the Electrical path that connects the CPU, memory & other hardware to the motherboard
- The BUS is a parallel lines, no. of lines decides the speed
- Since each wire can transfer one bit at a time, an 8-wire bus can move eight bits at a time, which is a full byte.

Address BUS

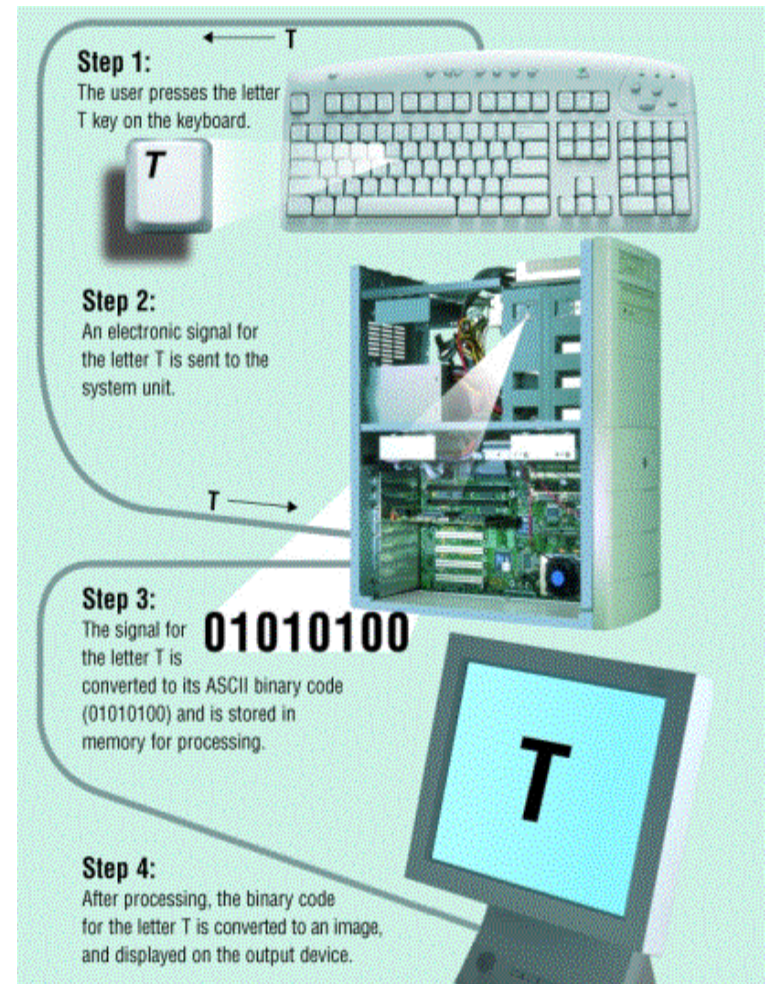
- Is a set of wires similar to the data bus, but it connects only the CPU and memory and all it carries are memory addresses

Computers process data by means of a binary system.

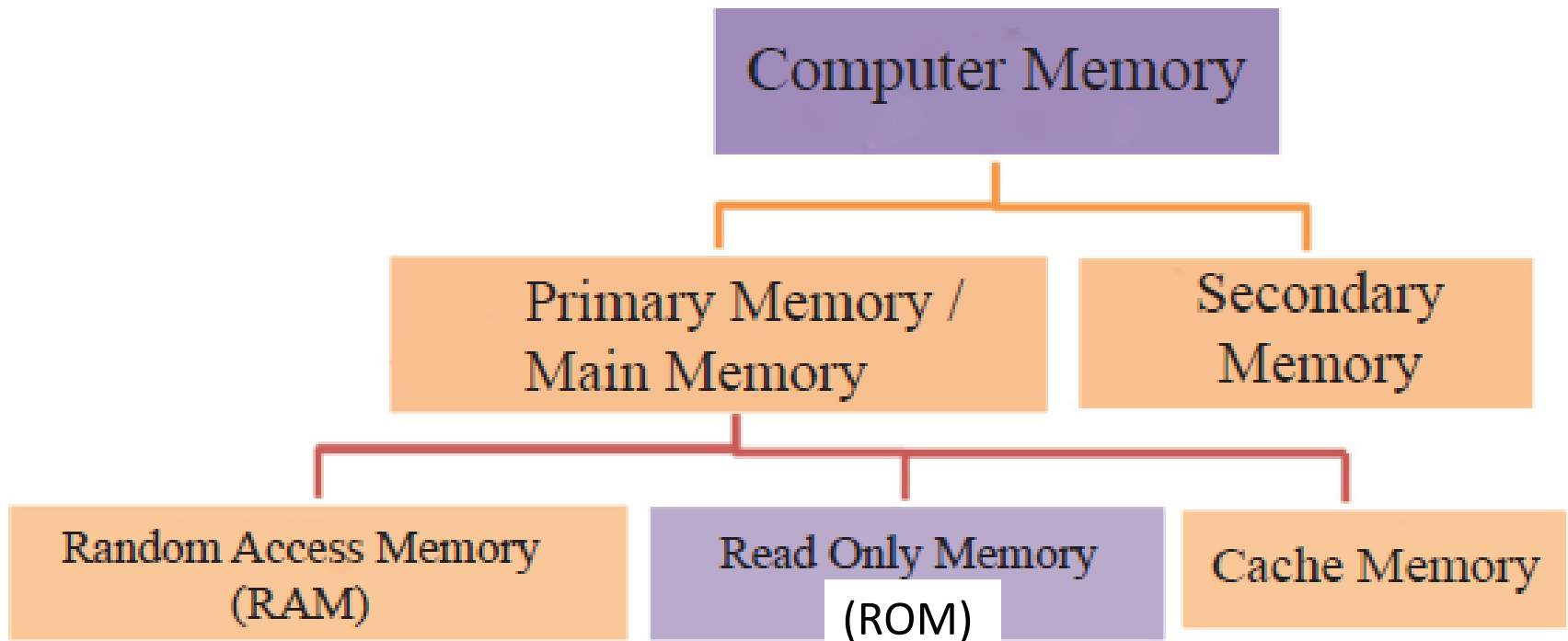
- A binary digit **1** or **0** is called a **bit**, the smallest unit of data in a computer system. Eight bits grouped together as a unit are called a **byte**, which provides enough combinations of 0s and 1s to represent 256 individual characters, including numbers, upper and lower case alphabet letters, punctuation marks and other characters.
- Coding schemes, such as ASCII, EBCDIC, and Unicode, provide the means to interact with a computer that recognizes only bits (on/off states).

If you press a letter on the keyboard...

When you press a letter on a keyboard, the electronic signals are converted into binary form and stored into memory. The computer then processes the data as bytes of information and converts them to the letters you see on the monitor screen or on a printed page.

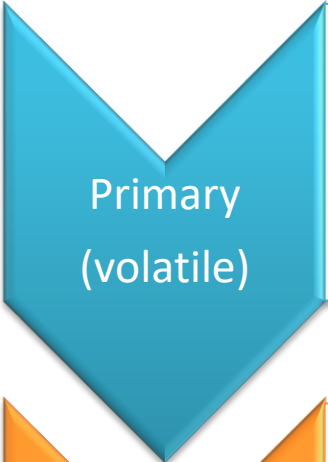


Computer Memory



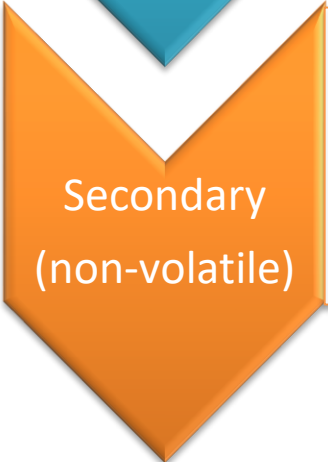
What is storage?

- Holds data, instructions, and information for future use



Primary
(volatile)

- Refers to main memory of a Computer, where data & instructions are held for immediate access and used by the Computer's CPU during processing this as a volatile memory

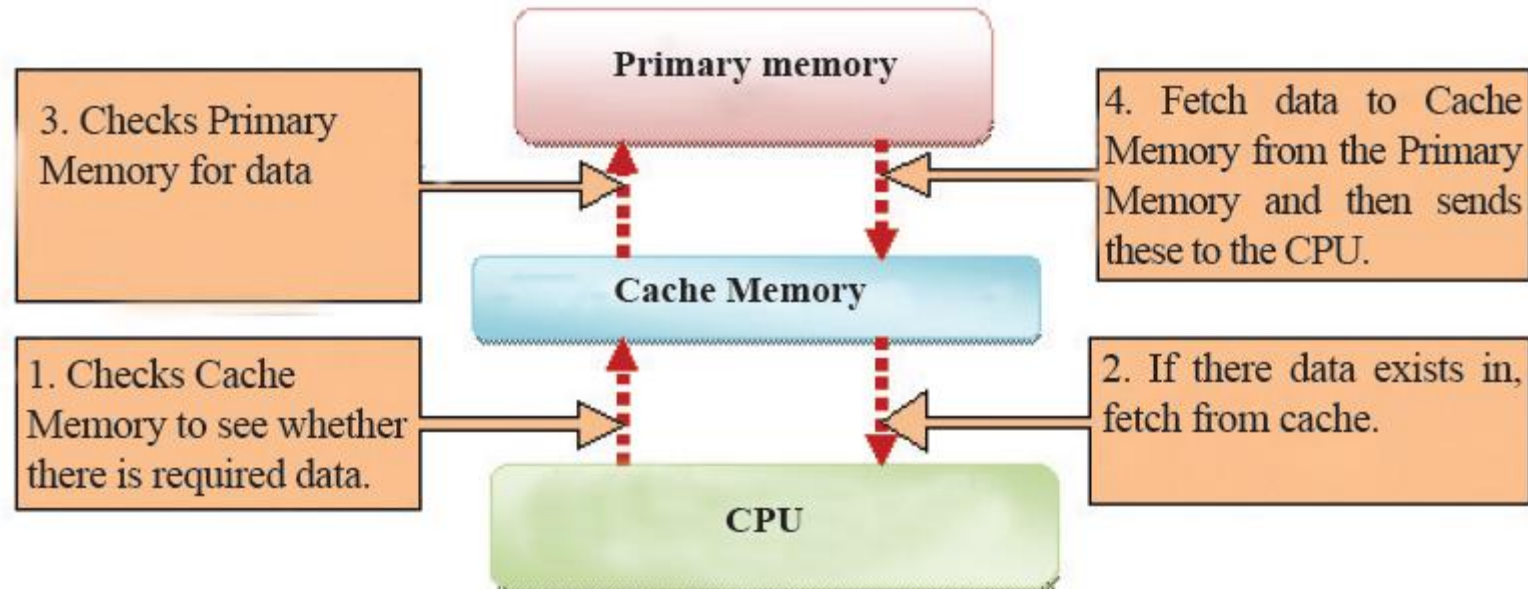


Secondary
(non-volatile)

- Is any storage devices designed to retain data and instruction in a more permanent form.
- Is non volatile

Cache Memory

- Though this is smaller in capacity than other memory, it is faster. This is also called CPU Memory.
- This acts as the mediator between CPU (Central Processing Unit) and Primary Memory.
- The data frequently used by are stored here.



How data is stored?

- To store data for later use two things are required.

Storage Medium

- Physical material on which data, instructions, and information are stored

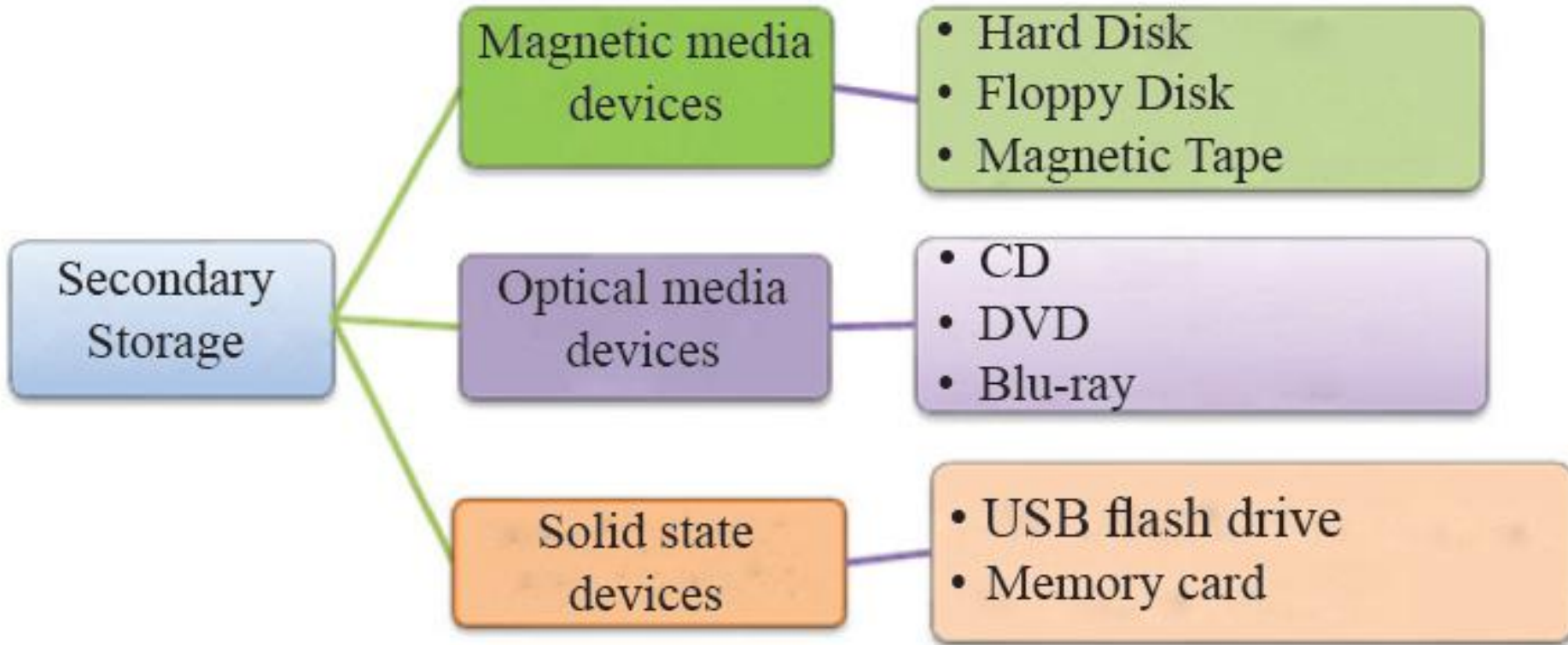
Storage Device

- Records and retrieves items to and from a storage medium

Storage Medium & Device

Storage Medium	Storage Device
Hard Disk (HD)	HD Drive
CD	CD Drive
External and Removable Hard Disks	USB port, FireWire Port, or other port on the system unit
Memory card miniature storage media	SD Slot(Secure Digital Slot)

Secondary Memory

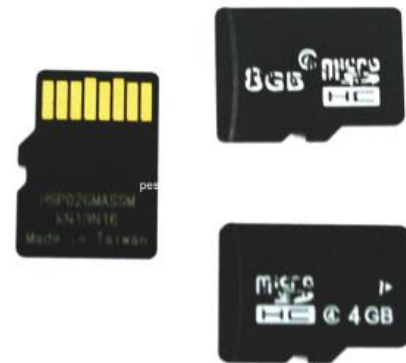


Optical Media Devices

Device	Type	Size	Image
CD-ROM (Compact Disc - Read Only Memory)	Used only to read data and information	650 – 900 MB	
CD-R (Compact Disc-Write once)	Can be used to write data and information Once until the capacity is filled.		
CD-RW (Compact Disc Re-Writable)	Data which is written on the disk can be erased and re-written.		
DVD-ROM Digital Versatile Disc – Read Only Memory	Used only to read data and information.	4.7 - 9.4 GB	
DVD-R (Digital Versatile Disc - Write Once)	Can be used to write data and information once until the capacity is filled.		
DVD – RW (Digital Versatile Disc - Rewritable)	Data which is written on the disk can be erased and re-written.		
Blue ray (BD) R (Recordable)	Can be used to write data and information once until the capacity is filled.	25 – 128 GB	
Blue-ray RE (Recordable Erasable)	More data can be re-written many times until the capacity is filled.		
Blue - ray 3D	Used to record video.		

Solid State Devices

- Since these devices contain solid parts and does not contain moving parts they are called Solid State devices.
- They are used to read and write data and information.



Storage Capacity measurement

- The **byte** is the unit of measure used to determine the capacity of storage devices used with any type of computer.

Bit	A binary digit (1 or 0)
8 Bits (1 character)	1 Byte
1kB	1024 Bytes
1MB	1024 kB
1GB	1024 MB
1TB	1024GB

For example: CD - 650MB

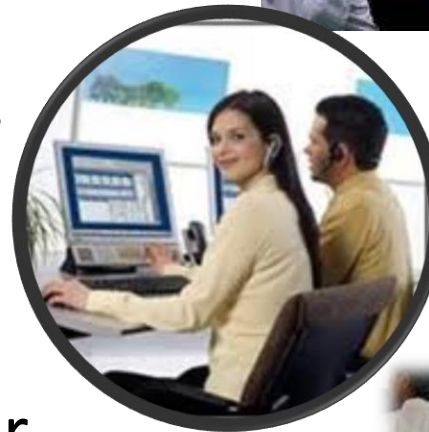
Live ware

- Live ware is the user who use the Computer
- Who is a **user**?
 - Person who communicates with a computer or uses the information it generates.



Live ware

- Examples for Live ware
 - System Operators
 - Computer Operators
 - Trainee / Student
 - Data entry operator
 - System Administrator
 - System developer & etc.

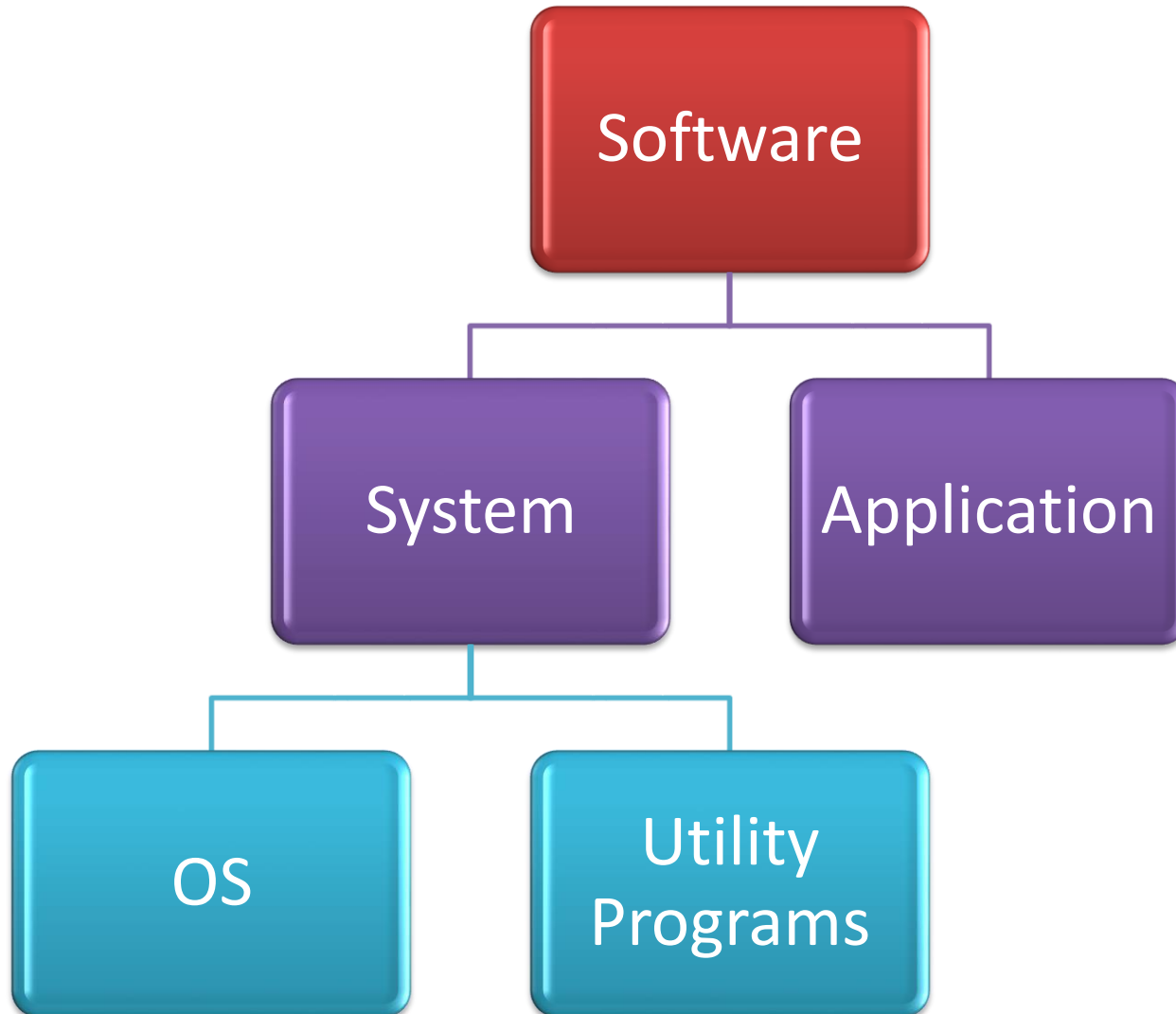


Computer Software

Computer Software

- Software is the term used to describe the instructions that tell the hardware how to perform a task.
- Without software instructions the hardware doesn't know what to do.

Computer Software



System Software

- Is the program a starts up the computer and functions on the principal of all hardware components and application software programs.
- Without system software loaded into the main memory of your computer your hardware and applications software will be useless.
- It includes OS, Utility programs, language translators & performance monitors.

Operating System (OS)

- An OS is itself a computer program.
- The OS wakes up the computer and makes the computer recognize the CPU, Memory, keyboard, video display system and disk drives.
- In addition it provides the facility for users to communicate with the computer and it serves as a platform from which to run application programs.

Operating System (OS)...

- OS has four primary jobs:

1

- Provide command line interface or GUI for the user to communicate with the computer

2

- Managing the hardware devices on the computer

3

- Managing and maintaining the disk

4

- Supporting other programs

Functions of OS

- The operating system's tasks, in the most general sense, fall into six categories:
 - Processor management
 - Memory management
 - Device management
 - Storage management
 - Application interface
 - User interface

Operating System (OS)...

- There are two categories of OS interfaces

Command Line Interface

- You type words and symbols using keyboard.

Graphical User Interface GUI

- With GUI you select actions by using mouse click on pictures, called icons or pick up options from menus
- Every OS provides GUI

Operating System (OS)...

Command Line Interface

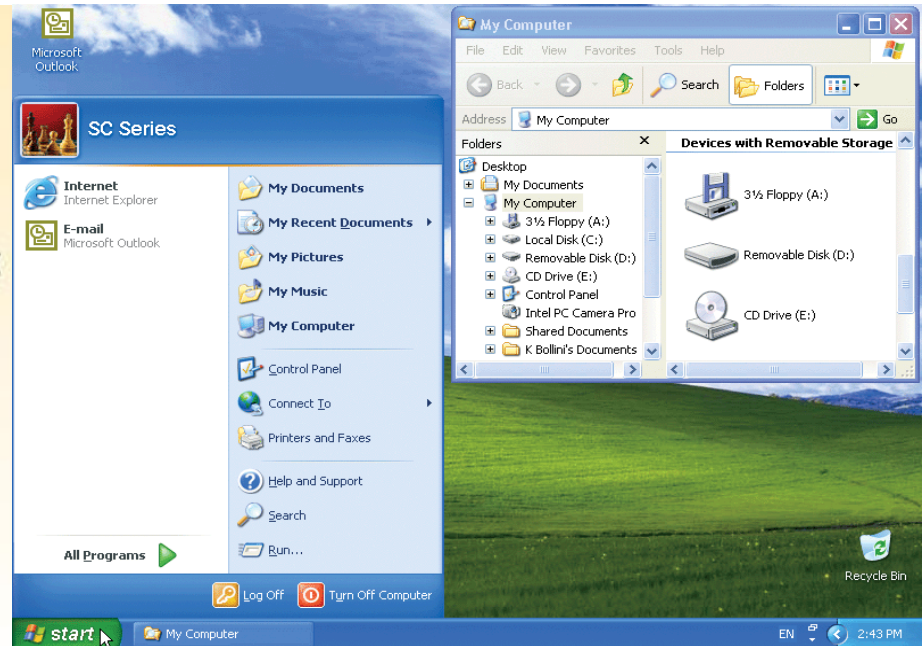
```
* COMPUTE REGULAR TIME PAY
MULTIPLY REGULAR-TIME-HOURS BY HOURLY-PAY-RATE
  GIVING REGULAR-TIME-PAY.

* COMPUTE OVERTIME PAY
IF OVERTIME-HOURS > 0
  COMPUTE OVERTIME-PAY = OVERTIME-HOURS * 1.5 * HOURLY-PAY-RATE
ELSE
  MOVE 0 TO OVERTIME-PAY.

* COMPUTE GROSS PAY
ADD REGULAR-TIME-PAY TO OVERTIME-PAY
  GIVING GROSS-PAY.

* PRINT GROSS PAY
MOVE GROSS-PAY TO GROSS-PAY-OUT.
WRITE REPORT-LINE-OUT FROM DETAIL-LINE
  AFTER ADVANCING 2 LINES.
```

Graphical User Interface GUI



Examples for System Software

OS

- Windows
- MAC OS
- LINUX
- SOLARIS

Utility Programs

- Virus guard
- Disk defragmenter
- Encryption Utility
- Backup software

Utility Programs

What is an **antivirus** program?

- A utility that prevents, detects, and removes viruses from a computer's memory or storage devices.



What happen when virus attack take place, how we can identify?

- Destroy or corrupt data
- Files and/or information disappear
- Computer may slowdown
- Peripheral devices are not function properly
- PC is acting strange ways -- from going totally dark to not launching programs
- The best way to know this is to use a good, updated antivirus program to scan your system thoroughly.

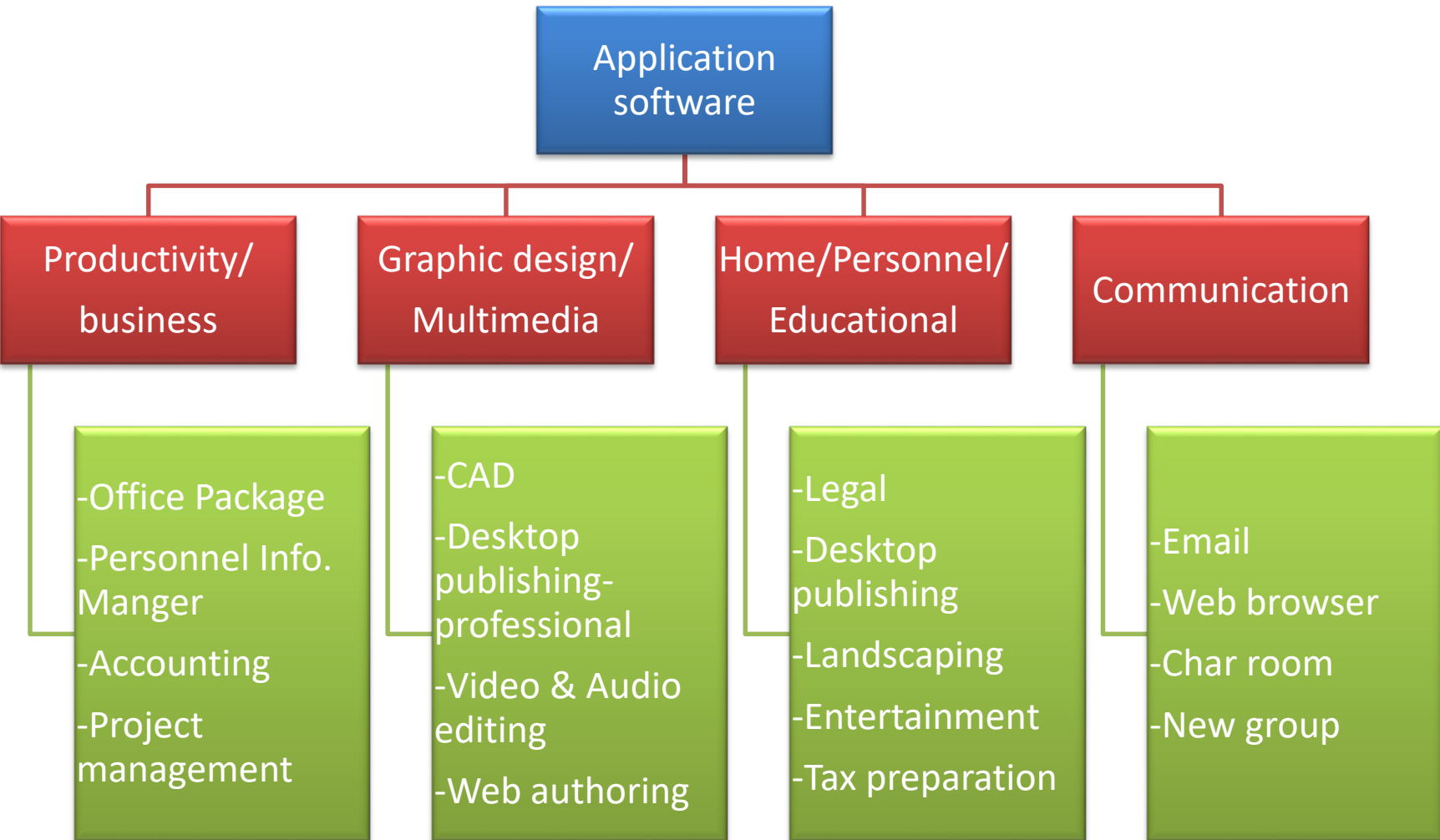
Application Software

- Programs that perform specific tasks for users
- Popular software applications bundled together as a single unit

Ex: Microsoft Office

Application Software

What are the categories of application software?



The Boot, or Startup, Process

The Boot Process, Step by Step

- Step 1: Power-on self test (POST)
- Step 2: The ROM BIOS startup program searches for and loads an OS
- Step 3: The OS configures the system and completes its own loading
- Step 4: The user executes applications software

What makes a computer powerful?



Disadvanges of using computers

- Depend on electricity
- Financial Frauds
- Abuses
- Environment Pollution
- Health issues

Computer Ergonomics

- Applied science devoted to comfort, efficiency and safety in work place
- Consideration should be given to the:
 - accessories required to operate properly
 - layout of equipment on the desk
 - location of furniture in the room.

Computer Ergonomics

- Keyboards

- Place the keyboard in a position that allows the forearms to be close to the horizontal and the wrists to be straight.

- Chairs

- Adjust the seat tilt so that you are comfortable when you are working on the keyboard.
- Your knees should be bent at a comfortable angle and greater than 90° flexion.

- Lighting

- Place the monitor to the side of the light source/s, not directly underneath. Try to site desks between rows of lights.

Computer Ergonomics

- Monitors

- Set the eye-to-screen distance at the distance that permits you to most easily focus on the screen. Usually this will be within an arm's length.
- Set the height of the monitor so that the top of the screen is below eye level and the bottom of the screen can be read without a marked inclination of the head.
- Usually this means that the center of the screen will need to be near shoulder height. Your eyes should be level with the tool bar.

Computer Ergonomics

- Using a mouse
 - A well-designed mouse should not cause undue pressure on the wrist and forearm muscles.
 - Keep the mouse as close as possible to the keyboard, elbow bent and close to the body.
- Posture while typing
 - Good posture is essential for all computer users. You should adopt a natural and relaxed position, providing opportunity for movement, from which you can assume a number of alternative positions.



Computer Ergonomics

